Lab 2: Agile Backlog & Sprint Simulation (Jira) - Coffee Kiosk

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Jira project: **Coffee-Kiosk** (key **CK**), Scrum template, company-managed. The Epics and User Stories below are derived from the Lab 1 functional and nonfunctional requirements. A ready-to-import file (**jira_import.csv**) with the same items is included in this folder.

1. Epics

Key	Epic	Description	Lab 1 requirements
E1	Epic 1: Customer Ordering Experience	Enable customers to browse, customise and place coffee orders through an intuitive self-service touch screen.	FR-001, FR-002
E2	Epic 2: Payment & Receipt	Let customers pay securely by credit card and receive a printed receipt that proves the purchase.	FR-003, FR-004, NFR-002
E3	Epic 3: Menu & Price Administration	Allow the cafe manager to maintain menu items and prices without developer help.	FR-005
E4	Epic 4: Kiosk Performance	Keep the kiosk fast enough to serve peak-hour queues.	NFR-001

2. User Stories (prioritised backlog)

Epic 1: Customer Ordering Experience

Story	As a	I want to	So that	Priority	Points	Sprint
Story 1.1: Coffee Selection	As a customer	I want to browse and select from Espresso, Americano and Latte	So that I can choose my preferred beverage	High	3	Sprint 1
Story 1.2: Size Selection	As a customer	I want to choose Small or Large and see the price for that size	So that I pay the right amount for the portion I want	High	2	Sprint 1
Story 1.3: Order Summary	As a customer	I want to review my coffee, size and total before paying	So that I can catch mistakes before I am charged	Medium	2	Sprint 1
Story 1.4: Cancel Order	As a customer	I want to cancel my order at any step before payment	So that I am never charged for an order I no longer want	Medium	1	Sprint 2

Epic 2: Payment & Receipt

Story	As a	I want to	So that	Priority	Points	Sprint
Story 2.1: Credit Card Payment	As a customer	I want to pay for my order with a credit card	So that I can complete my purchase at the kiosk	Highest	8	Sprint 1
Story 2.2: Declined Card Handling	As a customer	I want to be told when my card is declined and try another card once	So that I can still get my coffee	High	5	Sprint 2
Story 2.3: Printed Receipt	As a customer	I want to receive a printed receipt with my order number	So that I can collect my drink and keep proof of purchase	High	3	Sprint 1
Story 2.4: Secure Card Handling	As a cafe owner	I want card numbers and CVVs never to be stored on the kiosk	So that customers can trust the kiosk and we stay PCI-DSS compliant	High	5	Sprint 2

Epic 3: Menu & Price Administration

Story	As a	I want to	So that	Priority	Points	Sprint
Story 3.1: Admin Login	As a cafe manager	I want to log in to an admin screen with a PIN	So that only staff can change the menu	Medium	3	Sprint 2
Story 3.2: Update Prices	As a cafe manager	I want to edit the price of any menu item	So that the kiosk always shows current prices	Medium	3	Sprint 2
Story 3.3: Enable/Disable Items	As a cafe manager	I want to disable an item when it is out of stock	So that customers cannot order drinks we cannot make	Low	2	Backlog

Epic 4: Kiosk Performance

Story	As a	I want to	So that	Priority	Points	Sprint
Story 4.1: Fast Checkout	As a customer in a peak-hour queue	I want to finish my order in under 60 seconds	So that I do not hold up the people behind me	High	5	Backlog

3. Story Points (Fibonacci) and Estimation Rationale

Points use the Fibonacci scale (1, 2, 3, 5, 8, 13...) relative to the smallest story (1.4 Cancel Order = 1). Each story was estimated Planning-Poker style: pick a card for complexity, amount of work and uncertainty, then compare with already-estimated stories before settling on a value.

Story	Points	Why
1.1 Coffee Selection	3	Simple UI screen; menu data already defined.
1.2 Size Selection	2	Small change on top of 1.1; price lookup only.
1.3 Order Summary	2	Read-only screen reusing 1.1/1.2 data.
1.4 Cancel Order	1	One button plus state reset.
2.1 Credit Card Payment	8	External gateway integration; most uncertainty.
2.2 Declined Card Handling	5	Error paths and retry counter around 2.1.
2.3 Printed Receipt	3	Printer driver is known; layout work.
2.4 Secure Card Handling	5	Needs audit of DB, logs and receipts plus TLS checks.
3.1 Admin Login	3	Auth screen plus PIN storage.
3.2 Update Prices	3	CRUD form on the menu table.
3.3 Enable/Disable Items	2	Flag on existing menu rows.
4.1 Fast Checkout	5	Needs timing logs and UI tuning across all screens.

Total backlog: 42 points.

4. Sprint Plan (1-week sprints)

Sprint	Goal	Stories	Points
CK Sprint 1	A customer can pick a coffee and size, pay by card and get a receipt (happy path end-to-end).	1.1, 1.2, 1.3, 2.1, 2.3	18
CK Sprint 2	Handle payment failures, secure card data, cancellation, and give the manager price control.	1.4, 2.2, 2.4, 3.1, 3.2	17
Backlog	Carried to a later sprint.	3.3, 4.1	7

Workflow for each sprint: start sprint -> move stories To Do -> In Progress -> Done in priority order -> Complete sprint -> check Reports > Burndown Chart.

5. Screenshots

Saved in the **screenshots/** folder: (1) backlog with Epics and User Stories, (2) story point assignments, (3) active sprint board, (4) burndown chart for each sprint.

6. Reflection

1. Did your estimations reflect the actual effort? Mostly. The highest estimate went to 2.1 Credit Card Payment (8) because it depends on an external gateway; that was the item that stayed In Progress longest. Small UI stories (1.2, 1.3, 1.4) moved to Done quickly, which matched their 1-2 point estimates. Stories with hidden work, such as 2.4 Secure Card Handling, are the ones most likely to be underestimated, so a spike or a split would make the estimate more reliable.

2. Was your backlog well-prioritised? Yes. Sprint 1 contains only the stories needed for one complete paid order (select, size, summary, pay, receipt), so it delivers a usable kiosk on its own. Error handling, security hardening and admin features were placed in Sprint 2, and low-value items (3.3) stayed in the backlog.

3. How did your simulated sprint align with your plan? Both sprints were completed with all committed stories Done (18 points in Sprint 1, 17 points in Sprint 2). Because the sprint was simulated, stories were moved in priority order rather than over real working days, so the order of completion matched the plan exactly.

4. What insights did the burndown chart give about your team's capacity? The remaining-work line dropped in steps as each story was moved to Done. Large stories (8 points) cause big single drops, which shows that splitting them would give a smoother and more predictable burndown. The two sprints suggest a velocity of roughly 18 points per week, which is the figure to use when planning Sprint 3 (the remaining 7 points fit comfortably).